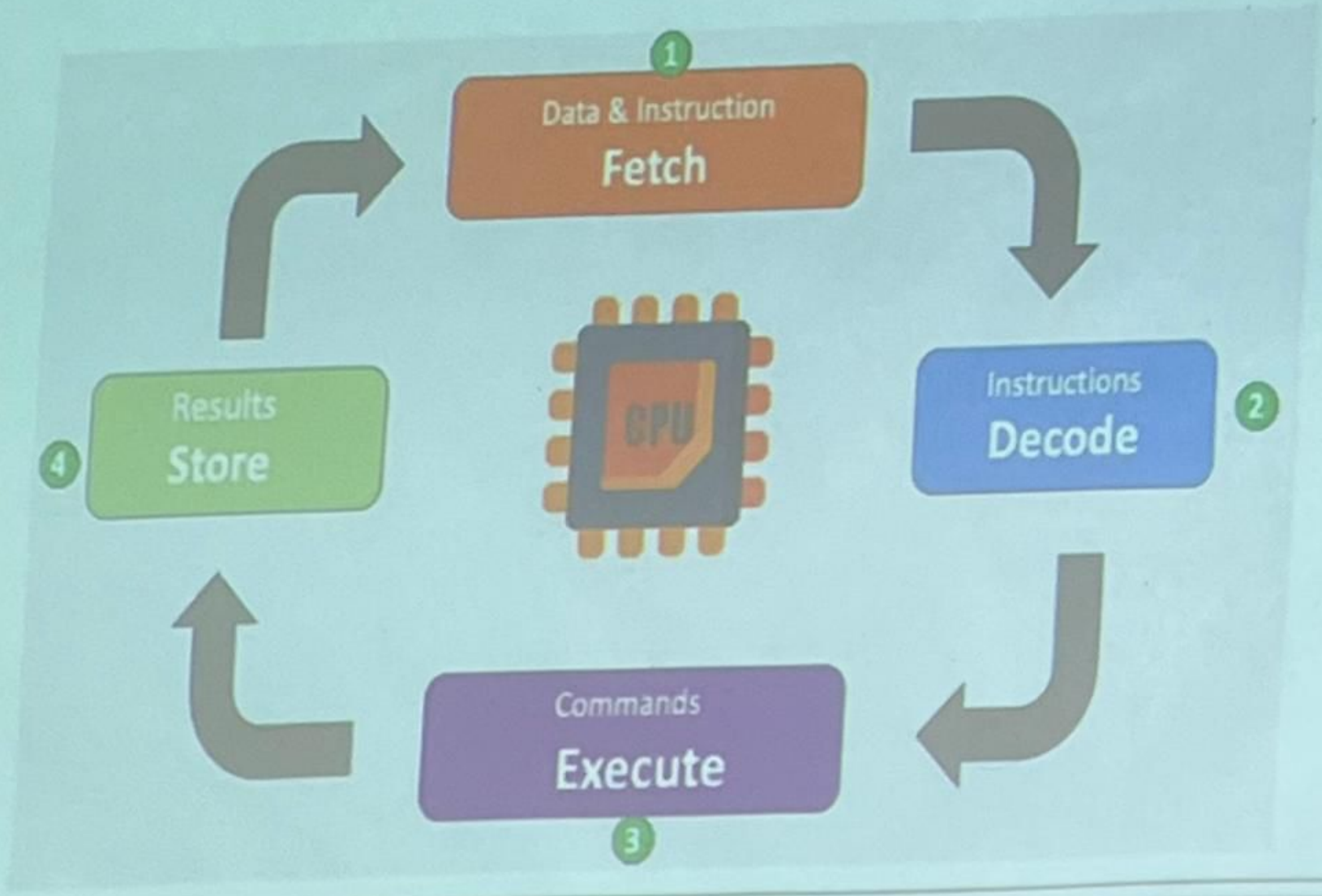
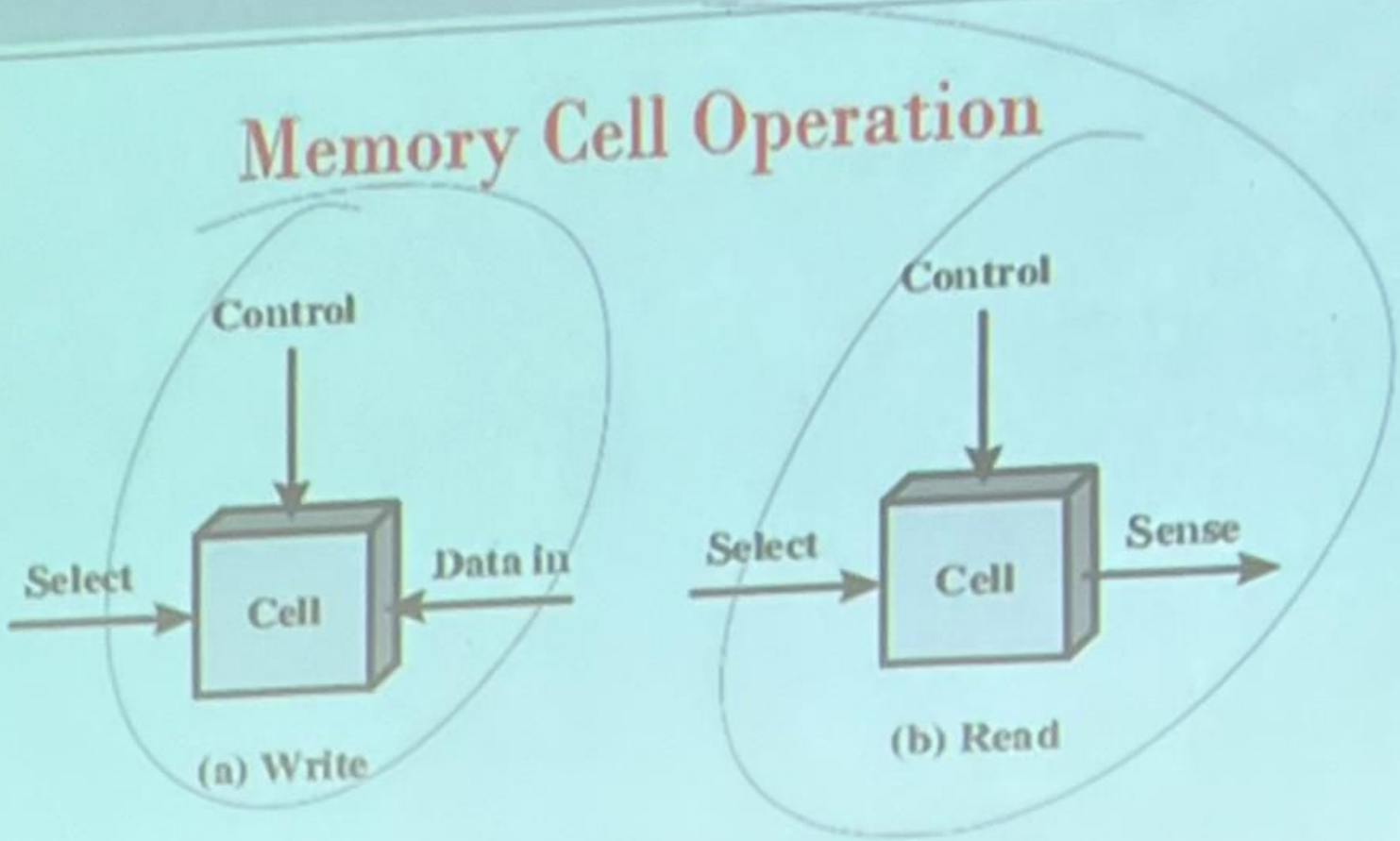
CPU Operation Phases

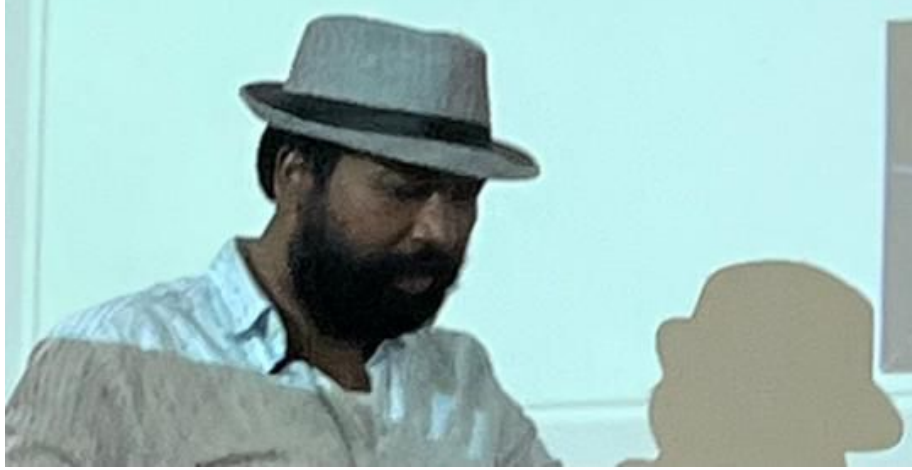


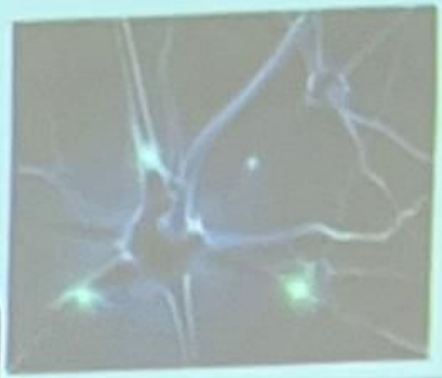
Memory

- *Computer memory is the storage space in the computer, where data is to be processed and instructions required for processing are stored.*
- The memory is divided into large number of small parts called cells.
- Each memory cell can hold one bit of information.
- Memory cells are organized in the form of an array.
- One row is one memory word.
- Passive component which stores data or instructions, accessed by address.
- Data flows from memory ("read") or to memory ("write").

Memory Cell Operation







Types of Memory: Location

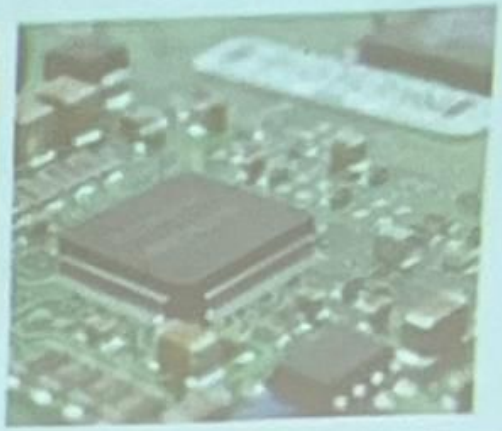
- The memory is that part of the computer where programs and data are stored.
- Computer memory system is divided into two parts:
  - **Internal memory**
  - **External memory**

Internal Memory Classification

- **According to Volatility:**
  - RAM
  - ROM
- **According to Technology:**
  - Static RAM
  - Dynamic RAM

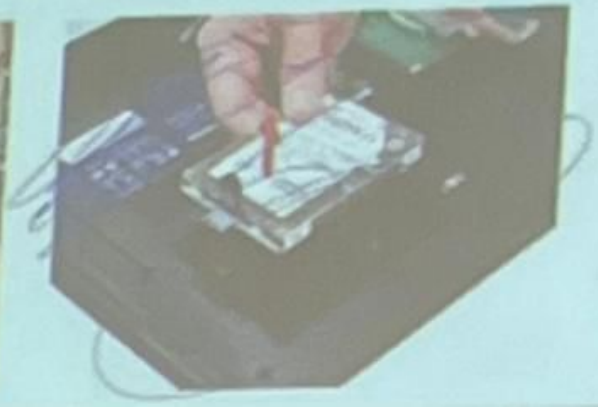
Internal Memory

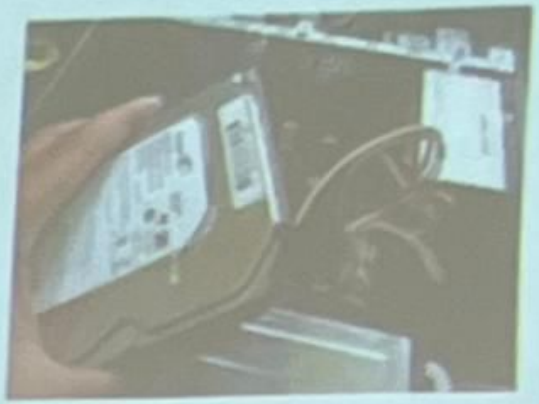
- Data / instruction stored inside this storage can be accessible easily by CPU.
- It's a part of main component of a computer system
- Types : register, cache memory and main memory



External Memory

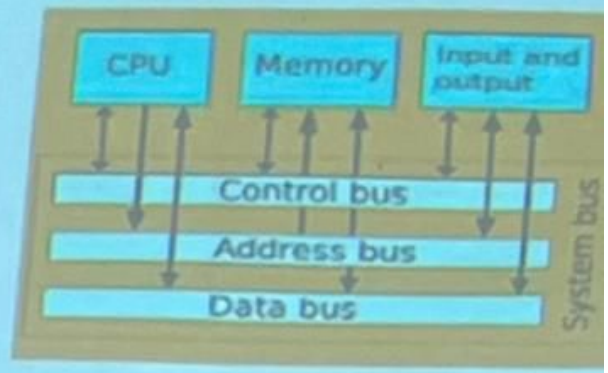
- Data / instruction stored inside this storage cannot be accessible easily by CPU, must go through I/O Module or I/O unit
- It's a not an integral part of main component of a computer system
- Examples: magnetic disks, optical disks etc.

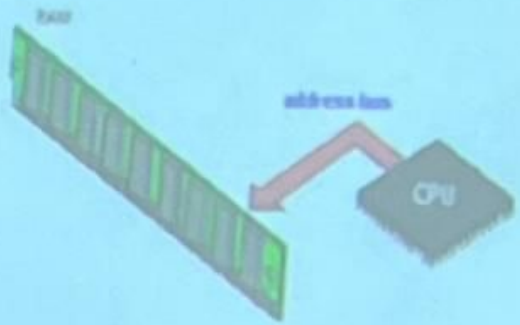




Primary Memory

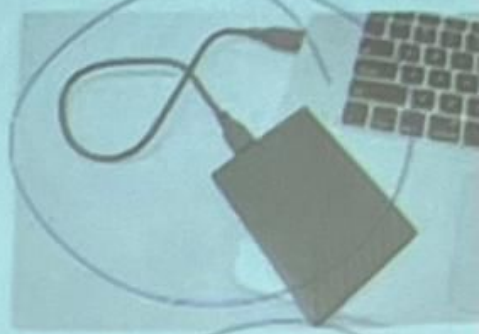
- Primary storage (or main memory or internal memory), often referred to simply as memory, is the only one directly accessible to the CPU.
- The CPU continuously reads instructions stored there and executes them as required.
- Main memory is directly or indirectly connected to the CPU via a *memory bus*.
- It is actually two buses (not on the diagram): an address bus and a data bus.
- The CPU firstly sends a number through an address bus, a number called memory address, that indicates the desired location of data.
- Then it reads or writes the data itself using the data bus.
- It is divided into RAM and ROM.





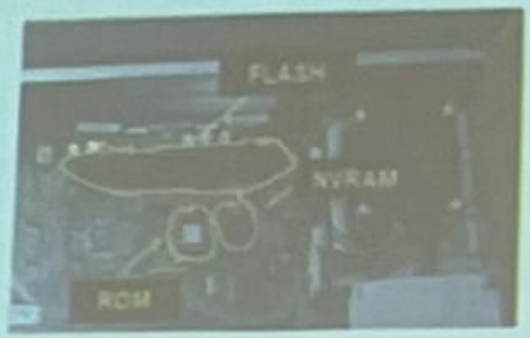
Secondary Memory

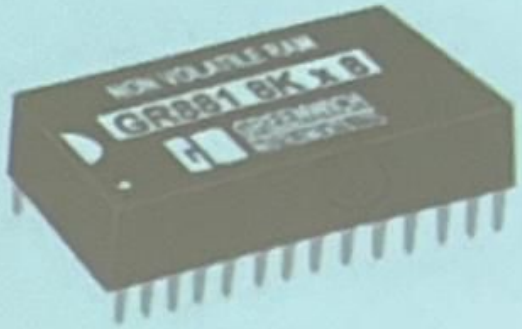
- The computer usually uses its input/output channels to access secondary storage and transfers the desired data using intermediate area in primary storage.
- Secondary storage does not lose the data when the device is powered down—it is non-volatile.
- Per unit, it is typically also an order of magnitude less expensive than primary storage.
- The secondary storage is often formatted according to a file system format, which provides the abstraction necessary to organize data into files and directories, providing also additional information (called metadata) describing the owner of a certain file, the access time, the access permissions, and other information.
- Hard disk are usually used as secondary storage.



Hybrid Memory

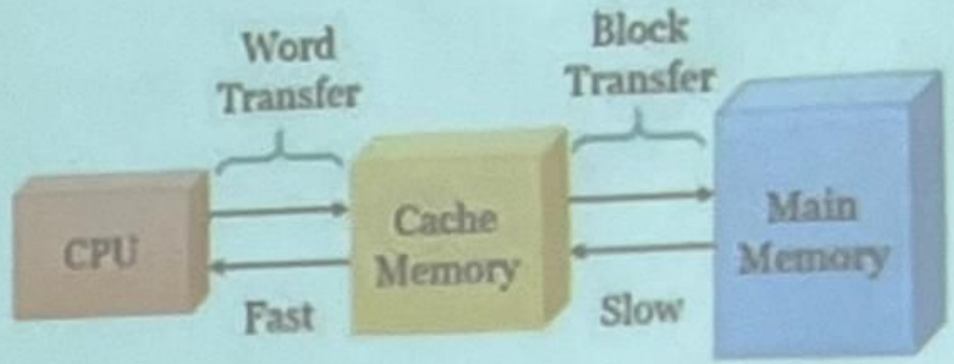
- As memory technology has matured in recent years, the line between RAM and ROM has blurred.
- Now, several types of memory combine features of both.
- These devices do not belong to either group and can be collectively referred to as hybrid memory devices.
- Hybrid memories can be read and written as desired, like RAM, but maintain their contents without electrical power, just like ROM.
- Two of the hybrid devices, EEPROM and flash, are descendants of ROM devices.
- These are typically used to store code. The third hybrid, NVRAM, is a modified version of SRAM.
- NVRAM usually holds persistent data.



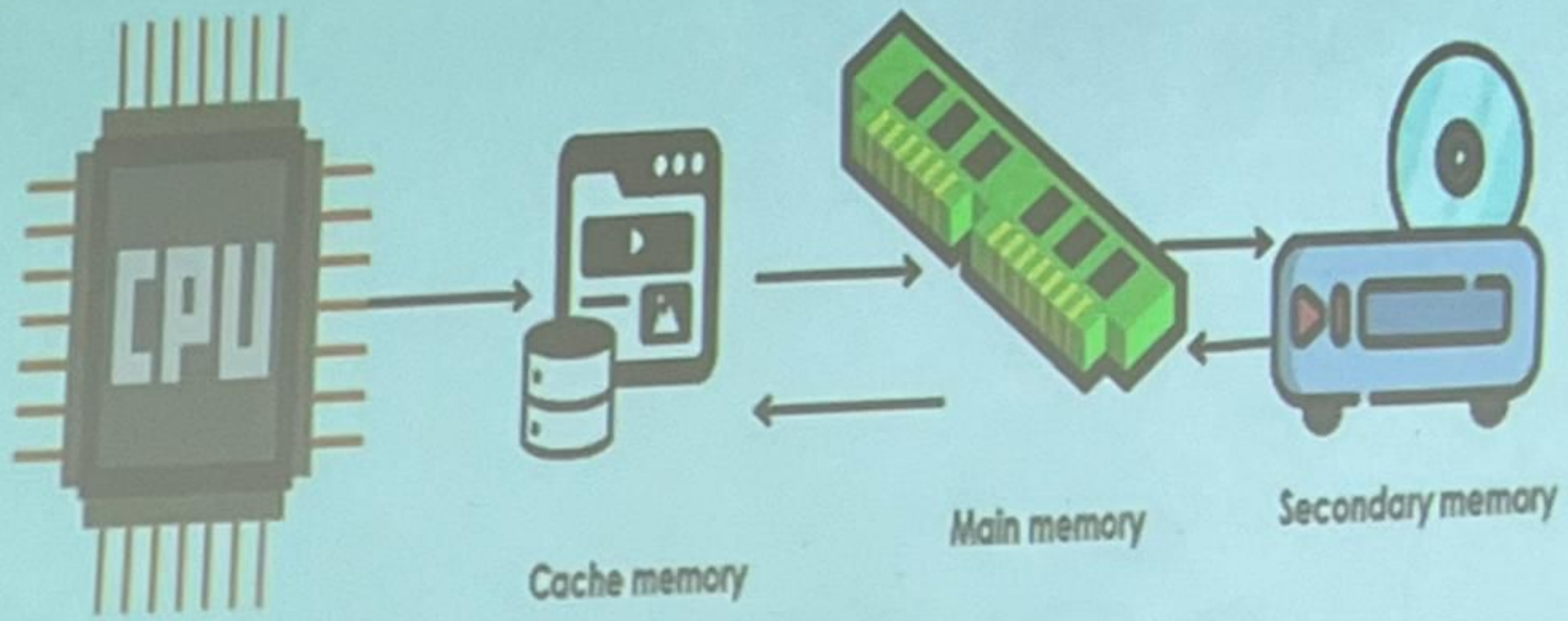


Cache Memory

- A CPU cache is a cache used by the central_processing unit of a computer to reduce the average time to access memory.
- The cache is a smaller, faster memory which stores copies of the data from the most frequently used main memory locations.
- As long as most memory accesses are cached memory locations, the average latency of memory accesses will be closer to the cache latency than to the latency of main memory.
- When the processor needs to read from or write to a location in main memory, it first checks whether a copy of that data is in the cache.
- If so, the processor immediately reads from or writes to the cache, which is much faster than reading from or writing to main memory

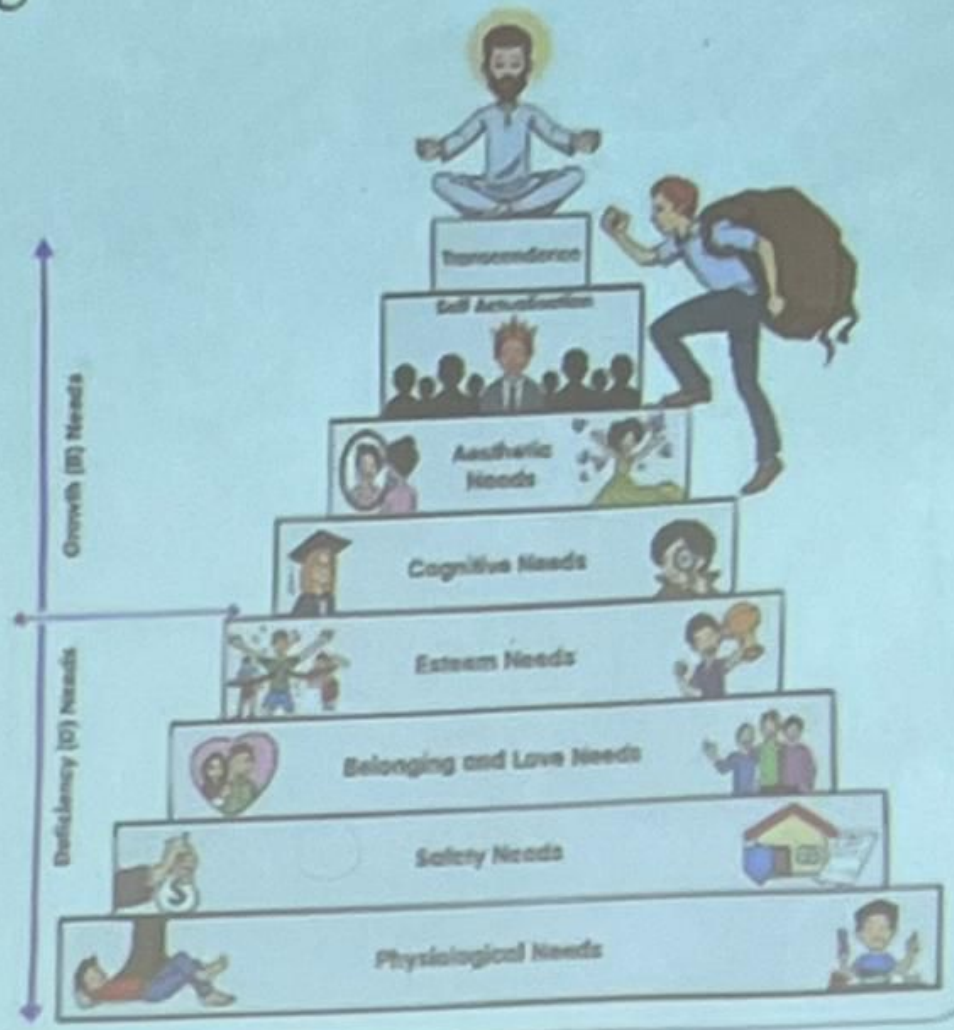


Different Memory



Memory Hierarchy

- *The Bottom Line:* there is a trade-off among the three key characteristics of memory.
  a) Capacity.
  b) Access time.
  c) Cost.



Memory Hierarchy

- The way out of this dilemma is not to rely on a single memory component or technology, but to employ a memory hierarchy.
- The design constraints on a computer's memory can be summed up by three questions:

a) **How much? → Capacity**

- ✓ The question of how much is somewhat open ended.
- ✓ If the capacity is there, applications will likely be developed to use it.

b) **How fast? → Time is money**

- ✓ The question of how fast is, in a sense, easier to answer.
- ✓ To achieve greatest performance, the memory must be able to keep up with the processor.
- ✓ As the processor is executing instructions, we would not want it to have to pause waiting for instructions or operands.

c) **How expensive? → Cost**

- ✓ For a practical system, the cost of memory must be reasonable in relationship to other components.

Memory Hierarchy

- A variety of technologies are used to implement memory systems, and across this spectrum of technologies, the following relationships hold:

a) *Faster access time, greater cost per bit*

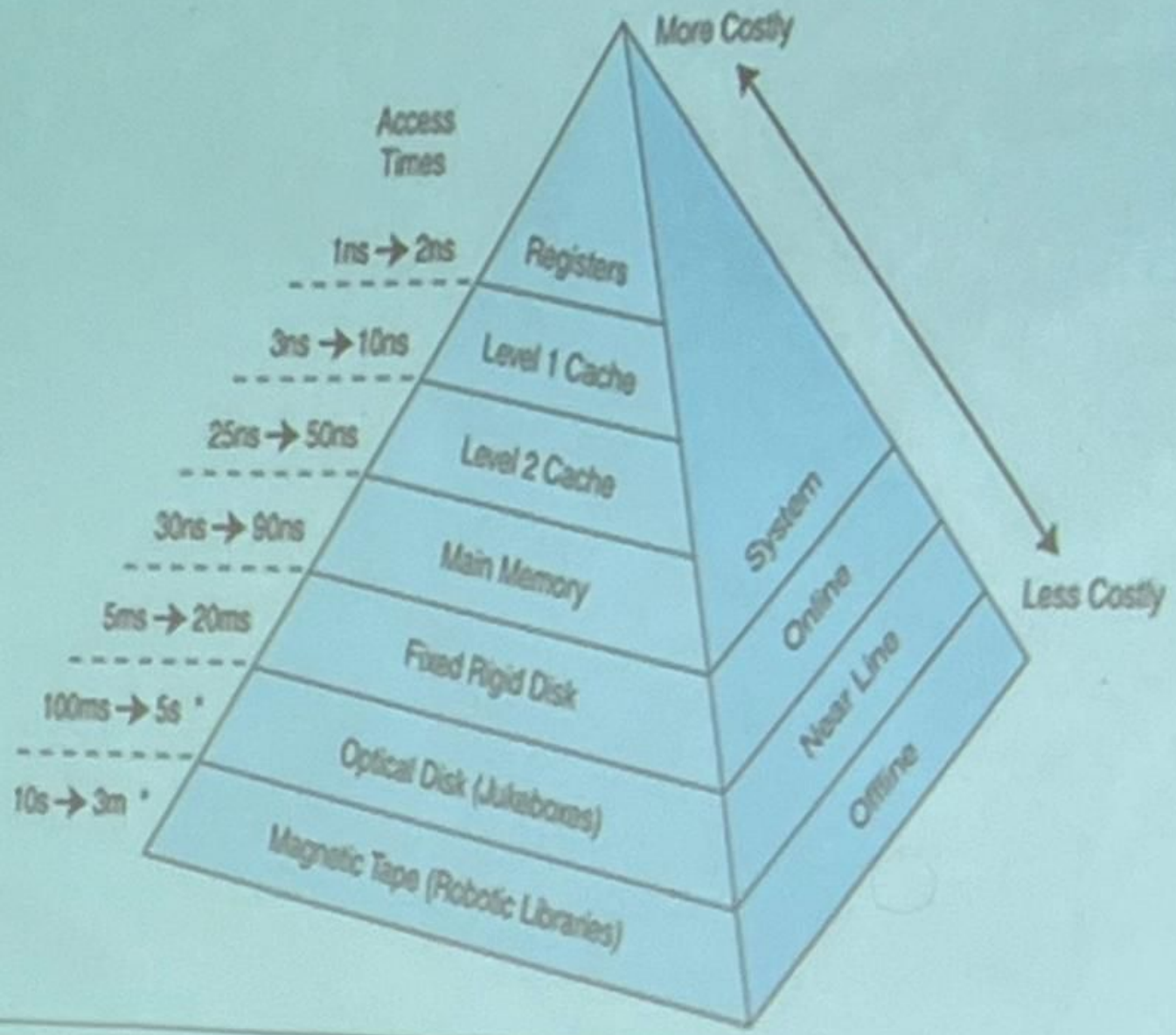
b) *Greater capacity, smaller cost per bit*

c) *Greater capacity, slower access time*

The designer would like to use memory technologies that provide for large-capacity memory, both because the capacity is needed and because the cost per bit is low.

However, to meet performance requirements, the designer needs to use expensive, relatively lower-capacity memories with short access times.

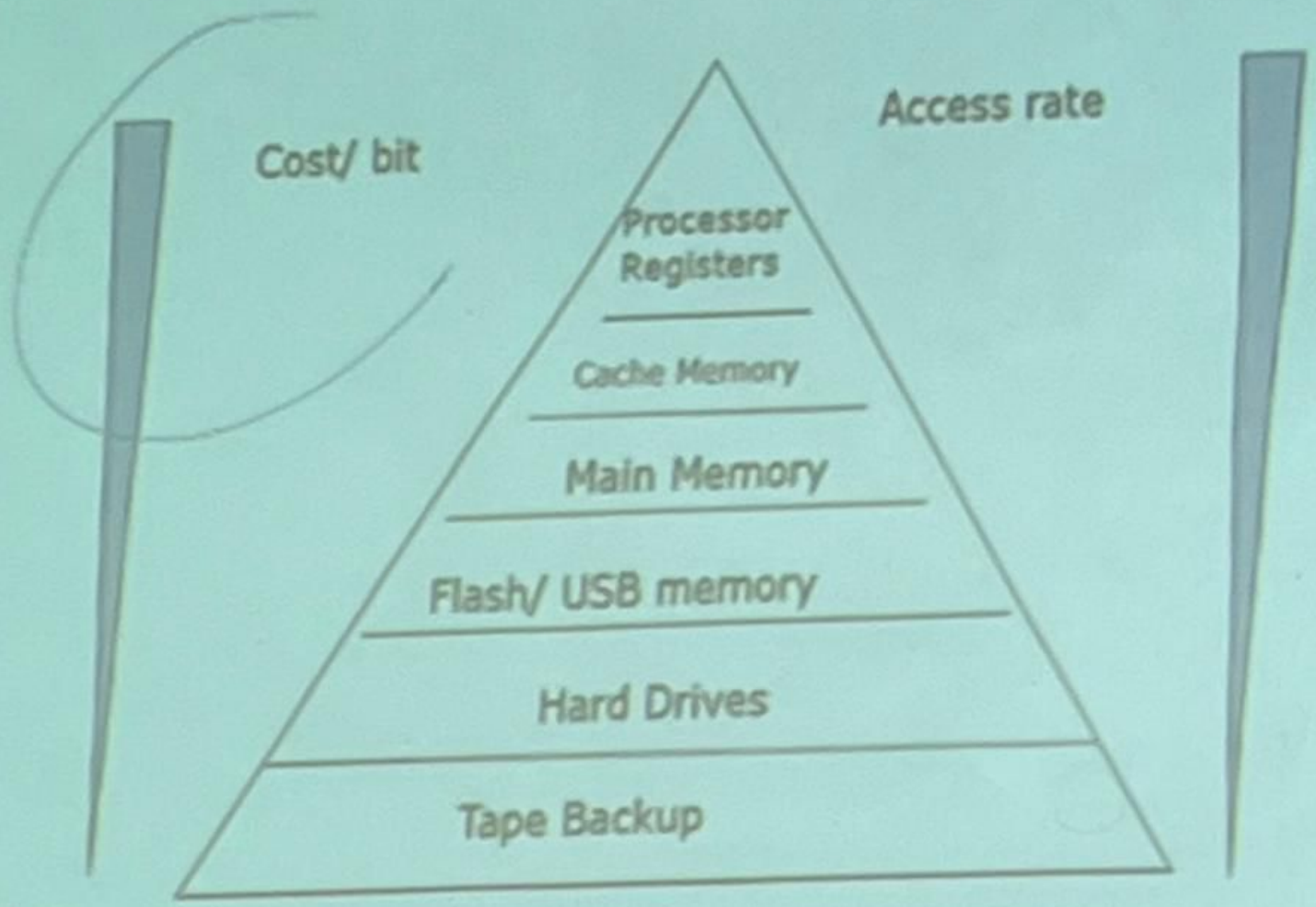
Memory Hierarchy- Pyramid



Memory Hierarchy

- As one goes down the hierarchy, the following occur:
  a) Decreasing cost per bit
  b) Increasing capacity
  c) Increasing access time
  d) Decreasing frequency of access of the memory by the processor

Memory Hierarchy



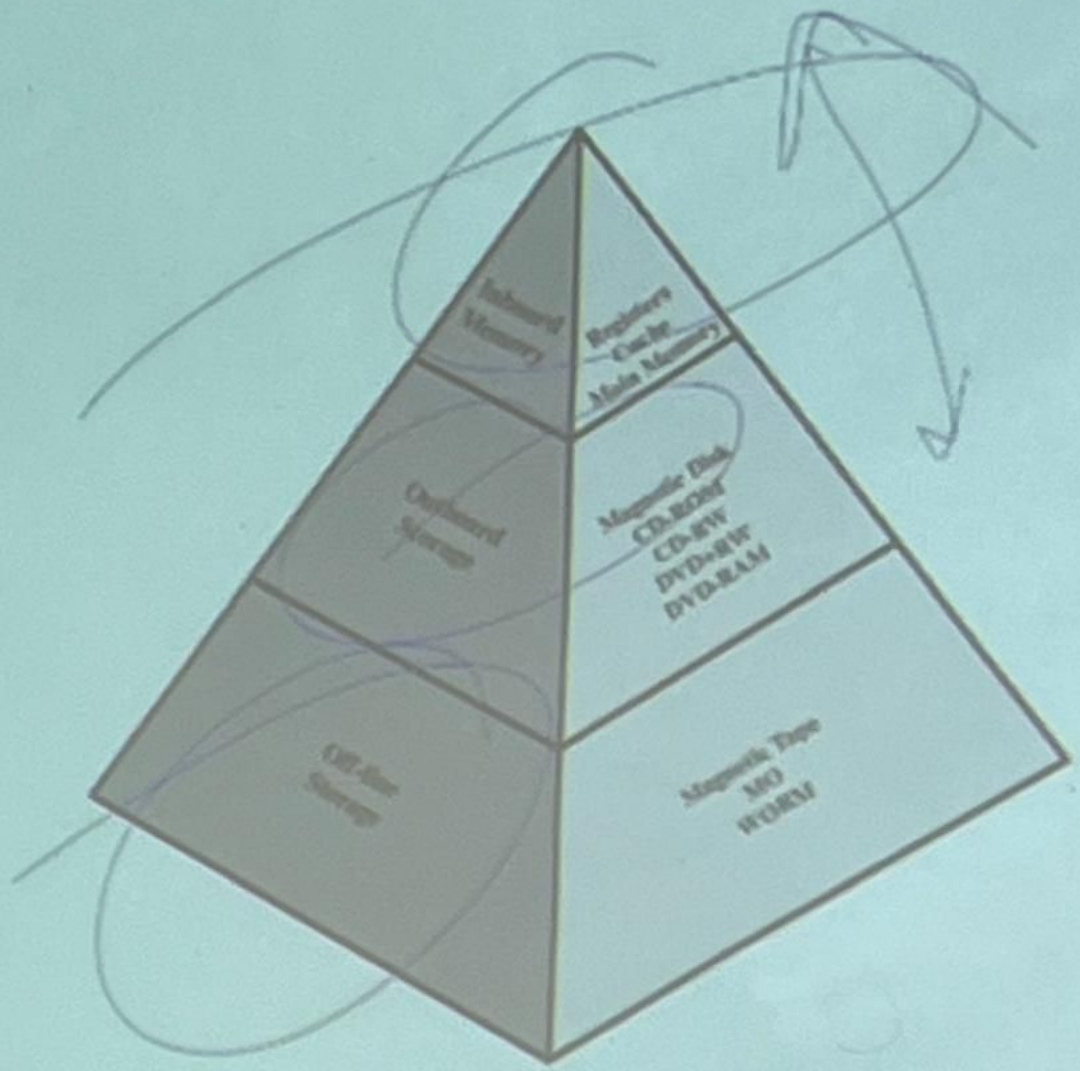
3-Level Memory Classification

A. **Inbound storage / memory**
   - Registers, Cache, Main memory.
   - The class of Internal memory

B. **Outbound Storage**
   - Magnetic Disk
   - CD ROM
   - CD RW
   - DVD + RW
   - DVD + RAM
   - The class of External memory

C. **Off-line Storage**
   - Magnetic Tape
   - Hard Disk Drive
   - The class of Backing store

Memory Hierarchy - Diagram



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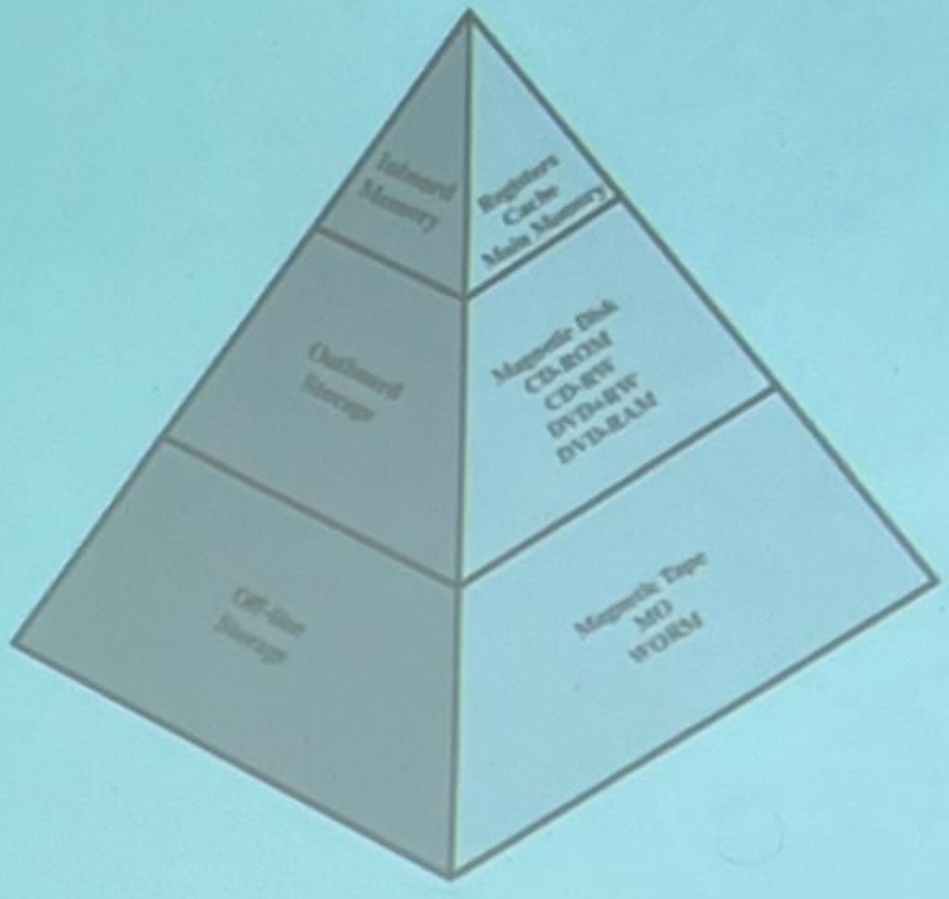
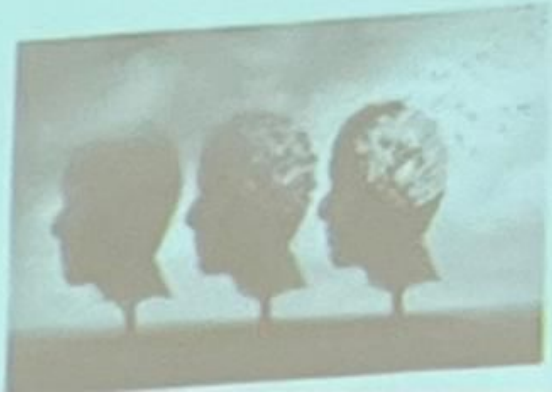
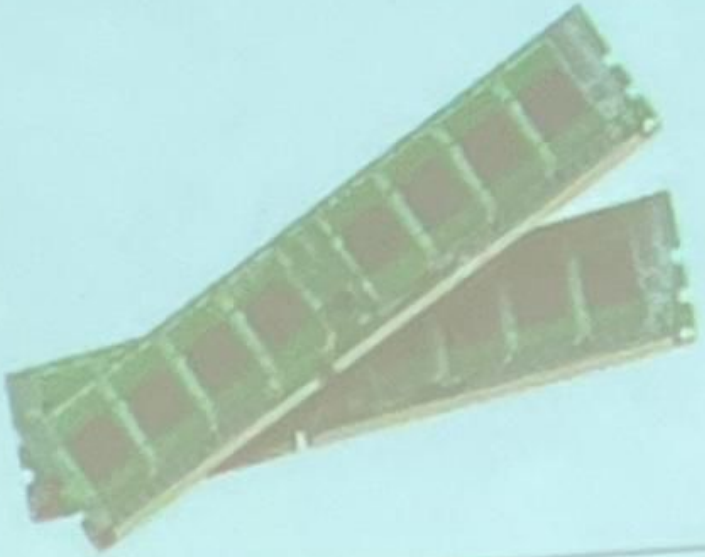
Memory Hierarchy - Diagram



Random Access Memory

- RAM is not a permanent storage place for information.
- It is active only when computer is on.
- If computer is switched off, the information is deleted from RAM.
- A temporary holding place for data and instructions.
- Consists of one or more chips on the motherboard.
- Sometimes called primary storage

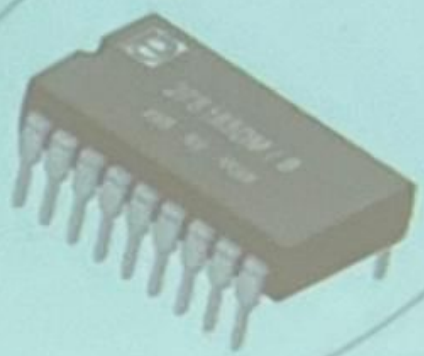


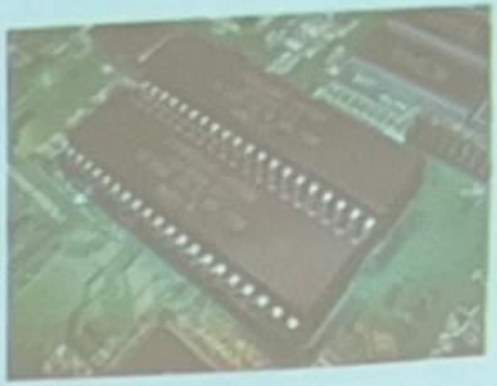


Read Only Memory

- ROM is the memory where information is only read.
- When the computer is turned a off, information is not deleted from ROM.

- Writing information to ROM is made by vendor.
  - Mostly used for installation into toys or appliances
  - Programmed once although there are exceptions
    - PROM/EPROM





Types of ROM

- Written during manufacture
  - Very expensive for small runs
- Programmable (once)
  - PROM
  - Needs special equipment to program
- Read "mostly"
  - Erasable Programmable (EPROM)
    - Erased by UV
  - Electrically Erasable (EEPROM)
    - Takes much longer to write than read
  - Flash memory
    - Erase whole memory electrically, no byte-level erase
    - But much faster write

Computer Memory